EXP.NO: 03
DATE: 23/07/26

LEARNING AND ASSIGNMENT OF IP ADDRESS MANUALLY TO COMPUTERS.

Aim:

To understand the IP addressing scheme and manually add IP addresses to computer on network.

Theory:

1. An IP address is an unique numerical label assigned to each device connected to a computer network.
2. IPV_4 addresses are 32-bit numbers written in dotted-decimal notation.
3. IPV_6 addresses uses 128-bit address.
4. Manual IP assignment is used for server and network devices requiring a permanent address
5. Subnet mask determines the network and host portions.
6. DHCP assigns IP dynamically.

Procedure:

1. In windows open network and Sharing center → change adapter settings → Right click ~~load~~ local area → properties → IPv_4 → set IP manually.
2. Enter IP address: 192.168.1.10, Subnet Mask: 255.255.255.0, Default gateway: 192.168.1.1
3. Enter DNS Server: 8.8.8.8, click Ok.
4. In Linux: Use command 'sudo ip addr add 192.168.1.10/24 dev etho'.
5. To set gateway in linux 'sudo ip route add default via 192.168.1.1'

6. For persistent configuration in Linux, edit /etc/network/interfaces or use nmcli
7. Verify the configuration using 'ipconfig' or 'ip a'.
8. Ping the default gateway to confirm network connectivity.
9. Ping another computer on the same subnet to verify communication.

Code:

```
Sudo ip addr add 192.168.1.10/24 dev eth0
Sudo ip route add default via 192.168.1.1 ip a
show eth0
ping 192.168.1.1
nmcli con mod 'wired connection 1' ipv4 addresses
                                    192.168.1.10/24
nmcli con mod 'wired connection1' ip v4 gateway
                                    192.168.1.1
nmcli con mod 'wired connection1' ip v4 method
nmcli con up 'wired connection1' manual
```

Result:

Static IP addresses were successfully assigned to computers. Connectivity was verified using ping commands between hosts on the same subnet.